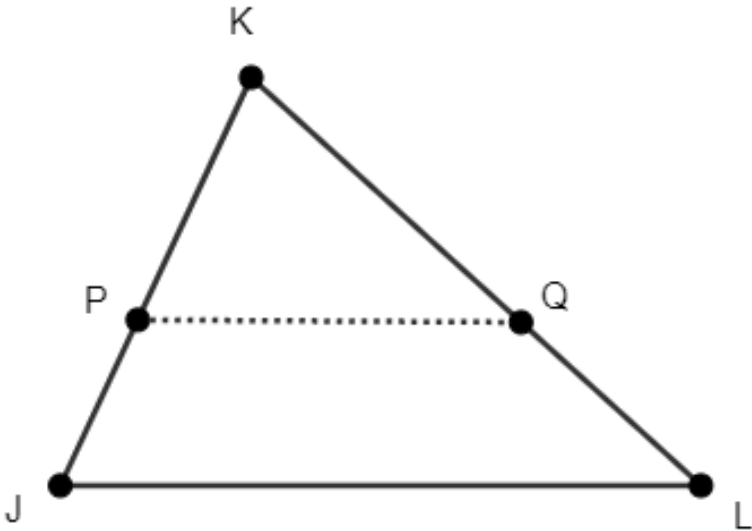


$\triangle KJL$ is shown, where $m\angle KJL = 65^\circ$, $m\angle KLJ = 42^\circ$, and $m\angle JKL = 73^\circ$. Point P is on $\overline{KJ}$ and Point Q is on $\overline{KL}$. $\overline{PQ}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle JKL$		Measurements of $\triangle PKQ$	
$JK =$	$m\angle KJL = 65^\circ$	$PK =$	$m\angle KPQ =$
$KL =$	$m\angle JKL = 73^\circ$	$KQ =$	$m\angle PKQ = 73^\circ$
$JL =$	$m\angle KLJ = 42^\circ$	$PQ =$	$m\angle KQP =$

2. The corresponding angles of $\triangle KJL$ and $\triangle KPQ$ are _____.

3. Complete the table.

Sides	Ratio
Determine the ratio of JK to PK .	
Determine the ratio of KL to KQ .	
Determine the ratio of JL to PQ .	

4. The corresponding sides of $\triangle KJL$ and $\triangle KPQ$ are _____.

5. $\triangle KJL$ is similar to $\triangle$ _____.
6. Describe a sequence of transformations that can be used to map $\triangle KJL$ onto $\triangle KPQ$.

Complete each relationship for $\triangle KJL$ and $\triangle KPQ$.

7. $\frac{PK}{JL} = \frac{\quad}{\quad}$

8. $\frac{JK}{QP} = \frac{\quad}{\quad}$

9. $\frac{KQ}{KJ} = \frac{\quad}{\quad}$

10. Observing that $\frac{JL}{PQ} \neq \frac{LQ}{QK}$, your classmate Moriah claims that $\triangle KJL$ and $\triangle KPQ$ are not similar. If her logic is faulty, explain why.